Sumo Logic SIEM

Hands-On Deployment & Testing Report

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Environment: AWS EC2 Windows Server (EC2AMAZ-FVEJPUR)

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1. What is Sumo Logic

Sumo Logic is a cloud-based SIEM (Security Information and Event Management) and log management platform. It collects logs from servers, applications, and cloud services in one place, so a SOC analyst can search, monitor, and set up alerts on that data instead of manually checking each machine. For this assignment, I deployed a collector agent on my own Windows VM (an AWS EC2 instance) and pulled real logs into Sumo Logic to see how the platform actually works end to end.

2. Key Features

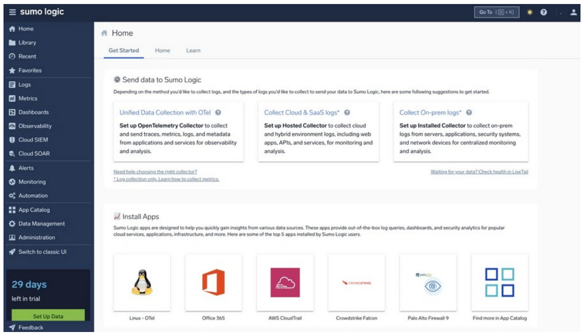
- Real-Time Monitoring — logs stream in as they're generated, viewable through Live Tail with near-zero delay.
- Threat Intelligence — built-in apps and rules that flag suspicious activity like repeated failed logins.
- Behavior Profiling — tracks baseline activity per host/user so unusual behaviour stands out.
- Data and User Monitoring — every log ingested carries metadata (host, source, collector) making it traceable back to the exact machine and user.
- Application Monitoring — supports app-specific log sources beyond just OS-level events (Docker, Host Metrics, Script sources, etc.).
- Analytics — Log Search lets you filter, aggregate, and visualize log data with a query language similar to Splunk's SPL.
- Log Management and Reporting — centralized storage with searchable retention, dashboards, and exportable results.

3. Deploying the Collector Agent

Goal: install a Sumo Logic Installed Collector on my Windows VM and get it registered and healthy.

Step 1 — Home | Step 2 — Add Installed Collector

Logged into the Sumo Logic console, then navigated to Data Management → Collection → Add Collector and picked Windows, since my VM runs Windows Server.



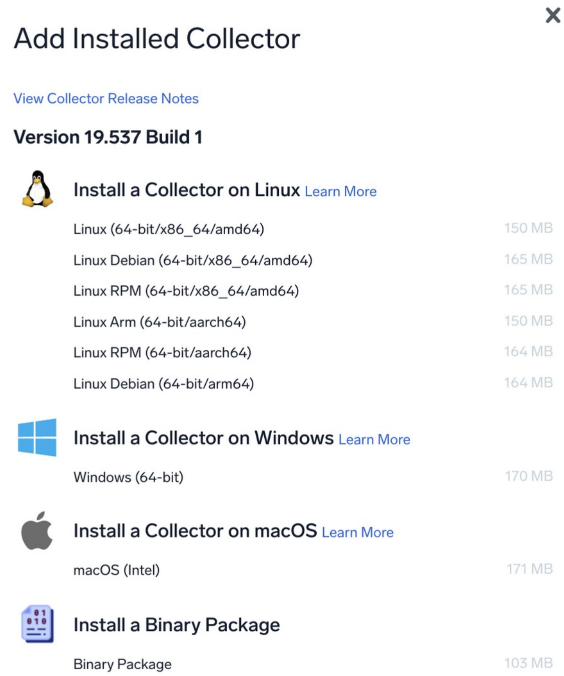
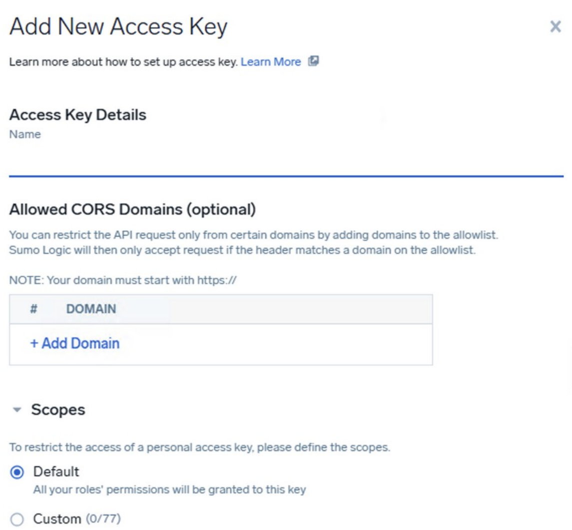


Fig 3.1 — Sumo Logic Home *Fig 3.2 — Add Installed Collector (Windows)*

Step 3 — Generate Access Key | Step 4 — Verify Collector is Healthy

Sumo Logic requires an Access ID/Key so the agent can authenticate back to my account. I created a new key, installed the Windows agent on the VM via RDP using that key, and confirmed the collector registered as Healthy.



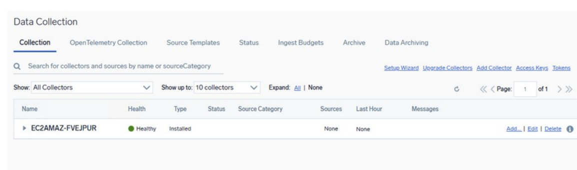


Fig 3.3 — New Access Key *Fig 3.4 — Collector Healthy*

4. Adding a Log Source & Verifying Logs Flow

A collector alone doesn't send data — it needs a Source attached. Sumo Logic offers several source types (File, Platform, and Windows-specific sources like Active Directory Inventory, Event Log, and Performance). I picked Windows Event Log, scoped to Security, Application, and System channels, then used Live Tail and Log Search to confirm data was arriving. The search for _sourceCategory=windows/security returned 149 results, including real EventCode values like 4624 (successful logon), 4634 (logoff), and 4672 (special privileges assigned).

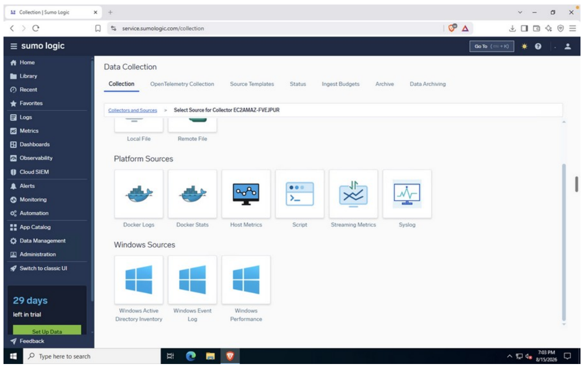


Fig 4.1 — Windows source types available

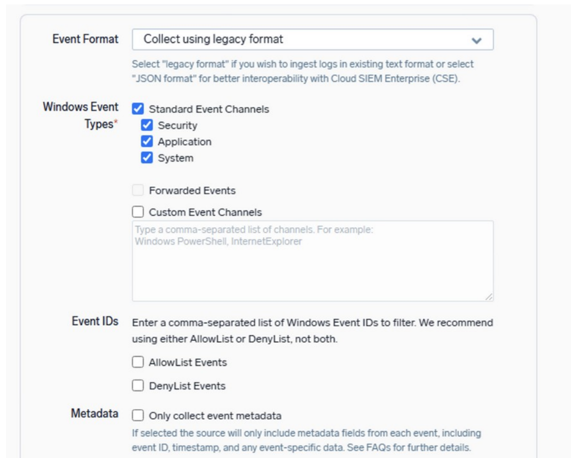


Fig 4.2 — Event source config

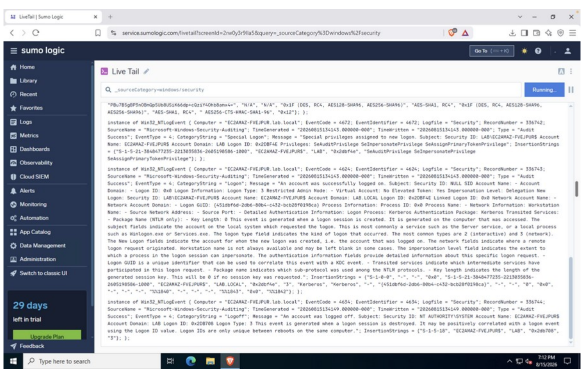


Fig 4.3 — Live Tail streaming events

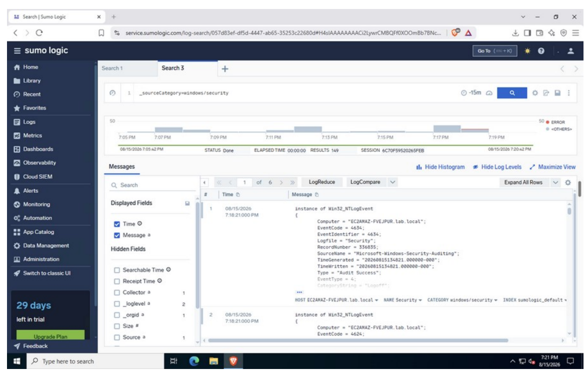


Fig 4.4 — Log Search, 149 results

5. Simulated Suspicious Activity Test

To test detection, I intentionally entered the wrong RDP password a few times before logging in correctly. Sumo Logic picked this up as EventCode 4625 (Audit Failure, failed logon) — exactly the kind of signal a SOC analyst would triage in a real environment.

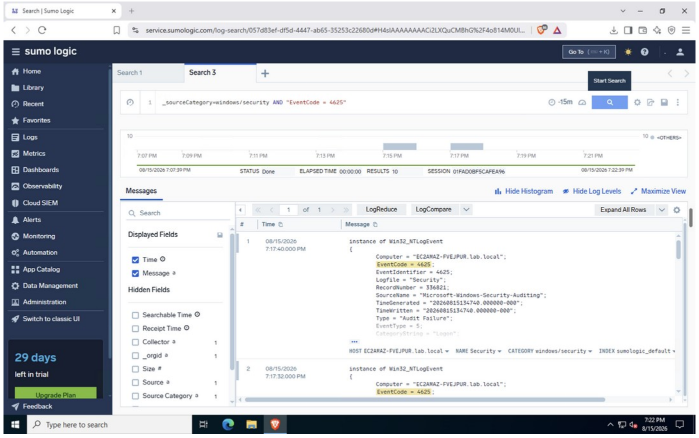


Fig 5.1 — 10 failed logon attempts (EventCode 4625) captured by Sumo Logic

6. Certification

Completed the Sumo Logic Academy self-paced onboarding track (learn.sumologic.com) covering data collection, search queries, and dashboard basics, as required for the tool certification.

7. Summary

By the end of this task I had a working Sumo Logic collector pulling live Windows Security logs from my AWS VM, confirmed the data was searchable, and verified that a real suspicious event (repeated failed logins) was captured and visible for triage. This gave me hands-on exposure to the core SIEM workflow — deploy collector, attach source, verify ingestion, search and detect — which is the same loop used in an actual SOC analyst role.